

BERKAY BENLI

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EDUCATION

Bachelor of Science (Honours) in Computing Science, University of Alberta (expected Winter 2027)

Relevant Courses

- **Object Oriented Programming, Data Structures and Algorithms, Programming in C**, Calculus I, Calculus II, Linear Algebra, **Computer Architecture, Algorithms I**

EXPERIENCE

University of Alberta

May 2024 – Aug. 2024

Research Assistant Intern

Edmonton, AB

- Developed and fine-tuned IMU data processing pipelines, leveraging **SciPy and Pandas** to filter signals and minimize noise.
- Implemented ROS scripts to integrate **IMU, LiDAR, and GNSS sensors**, enabling efficient data extraction and processing.
- Designed and deployed custom ROS packages for 3D vision using LiDAR, utilizing **Python, C++, PCL, and Open3D** to achieve high-precision indoor mapping with object segmentation and detection.

UASTarr Alberta Rocket Research

Jan 2024 - May 2024

Software Team Member

Edmonton, AB

- Collaborated with team members to **design and develop the UASTARR website**, addressing bugs, implementing new features, and **enhancing overall functionality** to improve user experience.
- Contributed to the **development of the rocket tracker software**, **enhancing tracking accuracy by 12%** through the implementation of more **efficient tracking algorithms** such as **Kalman Filter Algorithm**, and **optimizing code efficiency**

AlbertaSat

Sep. 2023 – Oct. 2023

Software Team Member

Edmonton, AB

- Contributed to the development of an **open-source React-based web application** with a **RESTful API backend**, designed for communication with a CubeSat satellite.
- Developed a C script leveraging the **Sub-GHz radio driver on an STM32 board** to receive and process radio packets transmitted from a team-built rocket.

PROJECTS

Sample-Searcher [App Link](#) — [Backend Code](#) — [Frontend Code](#) (React, NodeJS, Express, MySQL, Python, AWS)

- Developed a **full-stack** music application using React and NodeJS/Express with a MySQL database, integrating **Python web-scraped data** into a **MySQL database of over 2000 songs**.
- Deployed the backend on **AWS EC2** with **Docker containers** and implemented **CI/CD pipelines**, ensuring seamless updates and scalability.
- Created **RESTful API endpoints** for song retrieval and management. Designed a **responsive UI** with **dynamic song display** and pagination.

SKILLS

Computer Skills and Languages: Python, C, MySQL, JavaScript, React, NodeJS, HTML, CSS, Latex, Microsoft Office, Google Suite. Fluent in English and Turkish